

Lecture-#5

Natural Language Processing

A8706

Introduction

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

2. Finding structure of Documents

1. Introduction

- Finding the structure of documents is a crucial task in document analysis and information retrieval. Understanding the structure of a document allows for effective organization, navigation, summarization, and retrieval of information within the document. Here are some common techniques and approaches for finding the structure of documents:

Document Preprocessing:

Tokenization: Divide the text into individual tokens (words, phrases, sentences) to identify the basic units of text.

Sentence Segmentation: Split the text into sentences to analyze the document at a more granular level.

Headings and Sections:

Text Analysis: Use natural language processing (NLP) techniques to analyze the document's text and identify headings, subheadings, and section titles. These are often characterized by font size, style, or markup tags (e.g., HTML <h1>, <h2>).

Table of Contents: Extract the table of contents if available in the document.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

Tokenization

- **Sentence Tokenization:** Splitting the document into individual sentences.
- **Word Tokenization:** Splitting sentences into individual words or tokens.

Part-of-Speech Tagging (POS Tagging)

- Assigning parts of speech (e.g., noun, verb, adjective) to each word in a sentence. This helps in understanding the grammatical structure of the sentence.

Dependency Parsing

- Analyzing the grammatical structure of a sentence by identifying the dependencies between words. This method identifies which words are the subject, object, and how they relate to the main verb.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

Constituency Parsing

- Breaking down a sentence into sub-phrases or constituents, such as noun phrases (NP) and verb phrases (VP). This method helps in understanding how different parts of a sentence group together.

Named Entity Recognition (NER)

- Identifying entities (e.g., names of people, organizations, dates) in the text. NER helps in extracting important information and understanding the document's context.

Syntactic Tree Structures

- Representing the sentence structure as a tree where each node represents a syntactic category (e.g., NP, VP) and leaves represent words. Trees can be generated using parsing techniques like constituency or dependency parsing.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

Coreference Resolution

- Identifying when different words in a document refer to the same entity (e.g., "John" and "he"). This helps in maintaining coherence across the document.

Semantic Role Labeling (SRL)

- Assigning roles to words or phrases in a sentence to determine who did what to whom. SRL helps in understanding the sentence's meaning beyond just syntactic structure.

Text Structure Analysis

- For documents, analyzing the overall structure such as headings, paragraphs, and sections. This includes understanding the hierarchical nature of the document, identifying titles, abstracts, and main content sections.

Discourse Parsing

- Analyzing the structure of entire texts by identifying how sentences and paragraphs are connected logically. This includes understanding the rhetorical relations between different parts of the text.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

Tools and Libraries

NLTK (Natural Language Toolkit): A Python library for working with human language data.

spaCy: A powerful NLP library for parsing, tagging, and entity recognition.

Stanford NLP: A suite of NLP tools that includes parsers, POS taggers, and more.

CoreNLP: A tool from Stanford for natural language processing, including parsing and sentiment analysis.

OpenNLP: An Apache project that provides machine learning-based NLP tools.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

1. Sentence Boundary Detection

❖ **Sentence Boundary Detection (SBD)**, also known as sentence segmentation or sentence tokenization, is the process of identifying the boundaries between sentences within a block of text. Accurate sentence boundary detection is a crucial preprocessing step in natural language processing (NLP) tasks, as it allows you to analyze and process text at the sentence level.

❖ In written text in English and some other languages, the beginning of a sentence is usually marked with an uppercase letter, and the end of a sentence is explicitly marked with a period(.), a question mark(?), an exclamation mark(!), or another type of punctuation.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

- **Key Challenges in Sentence Boundary Detection**
- **Punctuation Ambiguity:**
 - Periods:** The most common sentence-ending punctuation mark can also appear in abbreviations (e.g., "Dr."), numbers (e.g., "3.14", "S.No."), or within quotes.
 - Ellipses:** "..." can indicate an unfinished thought or continuation rather than the end of a sentence.
 - Question and Exclamation Marks:** These are more reliable but still can be ambiguous in some contexts.
- **Abbreviations:**
 - Abbreviations (e.g., "e.g.," "i.e.," "Mr.") can falsely signal a sentence boundary when they appear at the end of a word.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

Quotation Marks:

Sentences ending with a quote can confuse the system as it may not be clear if the period or other punctuation inside the quotes ends the sentence or if the sentence continues outside the quotes.

Multiple Sentences in One Line:

Sometimes, multiple sentences are written in a single line without clear markers, making it harder to detect boundaries.

Languages with No Clear Sentence Boundaries:

In languages like Chinese or Japanese, where there is no punctuation to clearly mark sentence boundaries, SBD is particularly challenging.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

Rule-Based Approaches:

Punctuation Rules: One of the simplest methods is to split sentences at punctuation marks such as periods, question marks, and exclamation points. However, this method may lead to incorrect splits in cases where periods are used in abbreviations or numbers.

Abbreviation Detection: To address the issue of abbreviations, you can create rules to detect common abbreviations and avoid splitting sentences on periods within them.

Quotation Marks: Splitting sentences at quotation marks (e.g., ".", "?", or "!") outside of quotation marks can be effective for many texts.

Machine Learning Approaches:

Supervised Learning: You can train a supervised machine learning model (e.g., a binary classifier) to predict whether a punctuation mark indicates the end of a sentence. Features might include the punctuation mark itself, surrounding words, and contextual information.

Conditional Random Fields (CRF): CRFs are a type of probabilistic graphical model that can be used for sequence labeling tasks like SBD. They consider the entire sequence of tokens when making segmentation decisions.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

Deep Learning Approaches:

Recurrent Neural Networks (RNNs): RNNs, particularly Long Short-Term Memory (LSTM) or Gated Recurrent Unit (GRU) variants, can be used to model sequences of words and predict sentence boundaries based on the context.

Transformer Models: State-of-the-art models like BERT and GPT can be fine-tuned for sentence boundary detection as they have a strong understanding of contextual information.

Lexical Clues:

Capitalization: Sentences often start with a capital letter. You can use capitalization rules to detect sentence boundaries.

Numbers: Sentences often start after numbers, especially when followed by punctuation.

Language-Specific Approaches:

Languages may have specific grammatical rules that can be leveraged for SBD. **For example**, in languages like Japanese and Chinese, sentence boundaries are not indicated by spaces but by specific characters.

Third-Party Tools:

There are many NLP libraries and tools (e.g., NLTK, spaCy) that include pre-trained models and functions for sentence boundary detection. These tools often combine rule-based and machine learning techniques.

Evaluation:

Once you implement an SBD algorithm, it's important to evaluate its performance using metrics like precision, recall, and F1-score on a labeled dataset to ensure its accuracy.

Handling Edge Cases:

Consider handling edge cases, such as abbreviations, ellipses, and non-standard punctuation, to improve the robustness of your SBD algorithm.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

2. Topic Boundary Detection

- Topic boundary detection is a natural language processing (NLP) task that involves identifying points within a text where the topic or subject matter changes.
- Detecting topic boundaries is essential for various NLP applications, such as text summarization, content segmentation, document classification, and information retrieval. Here are some techniques and approaches for topic boundary detection:

Keyword-Based Approaches:

Identify keywords or key phrases that indicate a change in topic. These keywords can be domain-specific or general words that typically signal topic shifts (e.g., "however," "on the other hand," "in conclusion").

Term Frequency and Inverse Document Frequency (TF-IDF):

Calculate TF-IDF scores for terms in the document. A sudden increase in the TF-IDF score of a term may signal a topic boundary. A drop in TF-IDF values can also be indicative of a topic shift.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

2. Topic Boundary Detection

Cosine Similarity:

Measure the cosine similarity between document segments or paragraphs. A significant drop in similarity between consecutive segments may indicate a topic change.

Supervised Learning:

Train a supervised machine learning model, such as a binary classifier, to predict topic boundaries based on labeled data. Features for the model might include linguistic cues, such as punctuation, sentence length, and word usage.

Clustering:

Use clustering algorithms (e.g., k-means clustering) to group similar paragraphs or sentences together. Changes in cluster assignments can indicate topic boundaries.

Finding the Structure of Words and Documents



Learning from Experience, Sharing with Passion

•How TextTiling Works

•Tokenization:

–The text is first tokenized into words, and then into blocks (windows) of words.

•Lexical Scoring:

–The tokenizer calculates a lexical similarity score between adjacent blocks. If the similarity drops significantly, it's assumed that a topic boundary exists.

•Boundary Detection:

–The algorithm identifies where the largest drops in similarity occur and marks those points as potential topic boundaries.

